



# Advanced Time Series Analysis and Forecasting

## Seminar 10: Long memory and rough volatility



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## Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ **CEE**: Central and Eastern Europe
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **ELW**: Exact Local Whittle (estimator)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FCVAR**: Fractionally Cointegrated VAR
- ▣ **FFT**: Fast Fourier Transform
- ▣ **FIGARCH**: Fractionally Integrated GARCH

## Acronyms used in this material (2/2)

- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MSE**: Mean Squared Error
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **RFSV**: Rough Fractional Stochastic Volatility (model)
- ▣ **RV**: Realised Variance
- ▣ **SD**: Standard Deviation
- ▣ **SE**: Standard Error
- ▣ **VECM**: Vector Error Correction Model
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

## Today's question and seminar structure

- The question of the seminar
  - ▶ **how do we measure, test and use long memory and roughness in volatility and inflation?**
- Outputs to produce
  - ▶ the answers to the A exercises
  - ▶ one figure and one table for each B exercise
  - ▶ the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - ▶ one sentence of interpretation for each exercise
- Notebook for today: [open the seminar notebook in Google Colab](#)
  - ▶ each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, [danpele@ase.ro](mailto:danpele@ase.ro)

## Exercise map

| Task   | Question                                                                     | Type             | Model     |
|--------|------------------------------------------------------------------------------|------------------|-----------|
| A1, A2 | ARFIMA autocorrelations and the mean; fractional weights                     | Solved, Proposed | A1        |
| A3, A4 | standard errors of GPH and LW; bias and bandwidth                            | Solved, Proposed | A3        |
| A5, A6 | fractional Gaussian noise; the RFSV kernel                                   | Solved, Proposed | A5        |
| A7, A8 | circulant embedding by hand; switching that mimics long memory               | Solved, Proposed | A7, A1    |
| B1, B2 | memory of FTSE 100 log RV; BET and EUR/RON with the Qu test                  | Solved, Proposed | B1        |
| B3, B4 | roughness of the Euro Stoxx 50; Bitcoin by sampling interval                 | Solved, Proposed | B3        |
| B5, B6 | forecasting RV of the FTSE 100 and the CAC 40                                | Solved, Proposed | B5        |
| C1, C2 | fractional cointegration of two volatilities; what is wrong in an AI answer? | Proposed         | B1, A1–A8 |

- **[Solved]:** full solution in the slides and in the notebook, a model to follow; **[Proposed]:** you solve it, following the model; the solution is discussed in the seminar
- Part A: derivations and computations on paper; Part B: real data, each exercise ending with an interpretation question; Part C: open questions and the critique of an AI answer

## Data used

| Series                                     | Source                              | Frequency               | Period    |
|--------------------------------------------|-------------------------------------|-------------------------|-----------|
| RV of FTSE 100, Euro Stoxx 50, DAX, CAC 40 | <a href="#">Heber et al. (2009)</a> | daily, 5-minute returns | 2000–2022 |
| RV of Bitcoin                              | Binance one-minute prices           | daily, 1–60 minutes     | 2018–2026 |
| BET                                        | EODHD                               | daily                   | 2000–2026 |
| EUR/RON                                    | BNR reference rate                  | daily                   | 2005–2026 |

- ▣ RV in  $\%^2$ , returns in %
- ▣ Functions in the notebook
  - ▶ `gph`, `local_whittle`, `elw`, `qu_stat`, `gjr_scaling`, `h_with_noise`, `rfsv_kernel`, `forecast_rv`, `fcvar_fit`
- ▣ No account or key is needed

## Prerequisites for Today (1/24): autocorrelation and memory

- **Autocovariance** and **autocorrelation** of a stationary series  $X_t$  at lag  $k$

$$\gamma(k) = \text{Cov}(X_t, X_{t+k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)} \in [-1, 1]$$

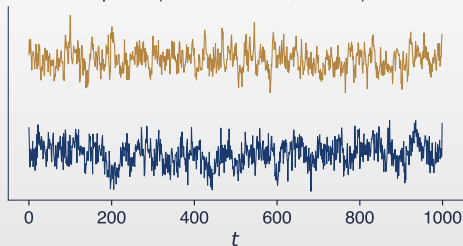
- ▶  $\gamma(0) = \text{Var}X_t$ ; stationary: mean and  $\gamma(k)$  do not depend on  $t$
- **Short memory**: autocorrelations decay exponentially and are summable
  - ▶ AR(1)  $X_t = \phi X_{t-1} + \varepsilon_t$ ,  $|\phi| < 1$ :  $\rho(k) = \phi^k$ , so  $\sum_k |\rho(k)| < \infty$
  - ▶  $\varepsilon_t$ : white noise (uncorrelated, mean 0, variance  $\sigma^2$ )
- **Long memory**: hyperbolic decay, not summable, a spectral pole at frequency zero

$$\rho(k) \sim c k^{2d-1} \ (k \rightarrow \infty), \quad f(\lambda) \sim G \lambda^{-2d} \ (\lambda \rightarrow 0), \quad 0 < d < 1/2$$

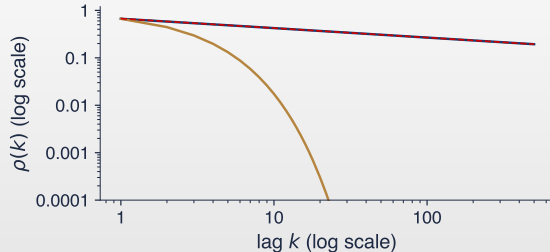
- ▶  $d$ : memory parameter;  $c, G > 0$ : constants;  $f(\lambda)$ : spectral density at frequency  $\lambda$  (defined below)
- ▶  $a_k \sim b_k$ : the ratio  $a_k/b_k \rightarrow 1$ ; Hurst exponent  $H = d + 1/2$

## Prerequisites for Today (2/24): hyperbolic against exponential decay

Simulated paths (same variance, shifted)



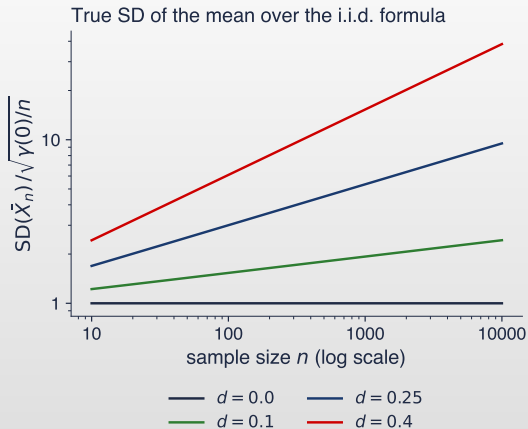
Autocorrelations



— AR(1),  $\phi = 0.67$     — ARFIMA(0, 0.4, 0)    - - -  $\frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$

- ARFIMA(0, 0.4, 0) (defined below) and an AR(1) with the same  $\rho(1)$ ,  $\phi = 0.67$ ; both simulated,  $n = 1000$
- At lag 100:  $\rho = 0.27$  for ARFIMA, practically 0 for the AR(1); on log scales the hyperbolic ACF is a straight line of slope  $2d - 1$
- The ARFIMA path wanders in long swings around its mean

## Prerequisites for Today (3/24): the variance of the sample mean



- For any stationary series

$$\text{Var}(\bar{X}_n) = \frac{1}{n} \sum_{|k| < n} \left(1 - \frac{|k|}{n}\right) \gamma(k)$$

- ▶  $\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t$ ;  $n$ : sample size
- ▶ i.i.d.: only  $k = 0$  remains,  $\gamma(0)/n$
- Long memory: of order  $n^{2d-1}$ , not  $n^{-1}$ 
  - ▶ the ratio in the chart grows like  $n^d$
  - ▶  $n = 1000$ : 1.9, 5.3, 15.3 for  $d = 0.1, 0.25, 0.4$
  - ▶ the i.i.d. standard error is that many times too small

## Prerequisites for Today (4/24): the lag operator and fractional differencing

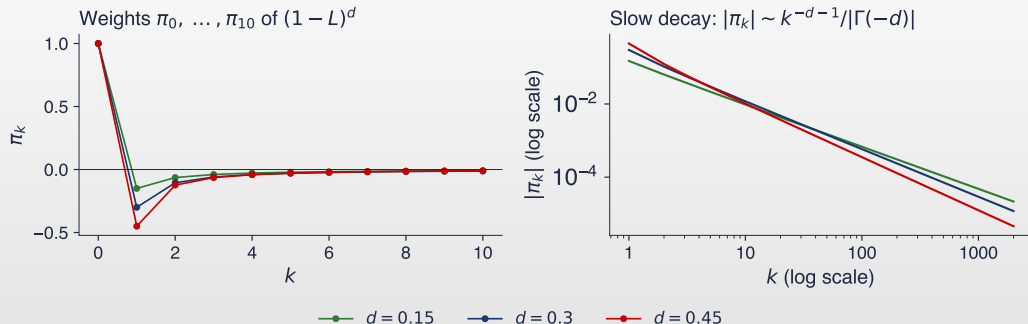
□ **Lag operator**  $L$ :  $LX_t = X_{t-1}$ ,  $L^k X_t = X_{t-k}$ ; first difference  $(1 - L)X_t = X_t - X_{t-1}$

□ **Fractional difference**: the binomial series of  $(1 - L)^d$  for a real  $d$

$$(1 - L)^d X_t = \sum_{k=0}^{\infty} \pi_k X_{t-k}, \quad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

- ▶  $\pi_k$ : weight of  $X_{t-k}$ ;  $d = 1$ :  $\pi_1 = -1$ ,  $\pi_k = 0$  for  $k \geq 2$  (the usual difference)
- ▶  $0 < d < 1$ : all past values enter, with weights  $\pi_k \sim k^{-d-1}/\Gamma(-d)$
- ▶  $\Gamma(z) = \int_0^{\infty} u^{z-1} e^{-u} du$ : gamma function,  $\Gamma(n) = (n-1)!$ ;  $\Gamma(-d) < 0$  for  $0 < d < 1$
- ▶  $\sum_k \pi_k = (1-1)^d = 0$ : a constant level is removed only by the infinite filter

## Prerequisites for Today (5/24): fractional weights



- Left:  $\pi_1 = -d$ , then small negative weights that shrink slowly
- Right: on log scales  $|\pi_k|$  is a straight line of slope  $-(d+1)$ : a hyperbolic, not exponential, decay
- A filter truncated after  $K$  lags leaves out the tail of the weights

## Prerequisites for Today (6/24): the ARFIMA model

- **ARFIMA**(0,  $d$ , 0) [Hosking \(1981\)](#): white noise after fractional differencing

$$(1 - L)^d X_t = \varepsilon_t, \quad -\frac{1}{2} < d < \frac{1}{2} \text{ (stationary and invertible)}$$

- ▶  $d > 0$ : long memory;  $d < 0$ : antipersistence ( $\sum_k \rho(k) = 0$ );  $d \geq 1/2$ : nonstationary

- Autocorrelations: a recursion and its limit

$$\rho(0) = 1, \quad \rho(k) = \rho(k-1) \frac{k-1+d}{k-d}, \quad \rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$$

- **ARFIMA**(1,  $d$ , 0):  $(1 - \phi L)(1 - L)^d X_t = \varepsilon_t$

- ▶  $\phi$ : AR coefficient,  $|\phi| < 1$ ; it changes the short lags,  $d$  the long ones
- ▶ **ARFIMA**( $p$ ,  $d$ ,  $q$ ): an **ARMA**( $p$ ,  $q$ ) for  $(1 - L)^d X_t$

## Prerequisites for Today (7/24): the spectral density and the periodogram

- **Spectral density:** the variance of  $X_t$  decomposed by frequency

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-i\lambda k}, \quad \gamma(0) = \int_{-\pi}^{\pi} f(\lambda) d\lambda$$

- ▶  $\lambda \in [-\pi, \pi]$ : angular frequency, a cycle of  $2\pi/\lambda$  periods; low  $\lambda$ : long cycles
- ▶ ARFIMA(0,  $d$ , 0):  $f(\lambda) = \frac{\sigma^2}{2\pi} (4 \sin^2(\lambda/2))^{-d} \sim \frac{\sigma^2}{2\pi} \lambda^{-2d}$

- **Periodogram** at the Fourier frequencies: the sample counterpart of  $f$

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^n X_t e^{-i\lambda_j t} \right|^2, \quad \lambda_j = \frac{2\pi j}{n}, \quad j = 1, \dots, \lfloor n/2 \rfloor$$

- ▶  $I(\lambda_j)/f(\lambda_j)$  is approximately exponential with mean 1, independent across  $j$ : very noisy
- ▶ bandwidth  $m$ : the number of lowest frequencies  $\lambda_1, \dots, \lambda_m$  used to estimate  $d$ ;  $m \rightarrow \infty$ ,  $m/n \rightarrow 0$

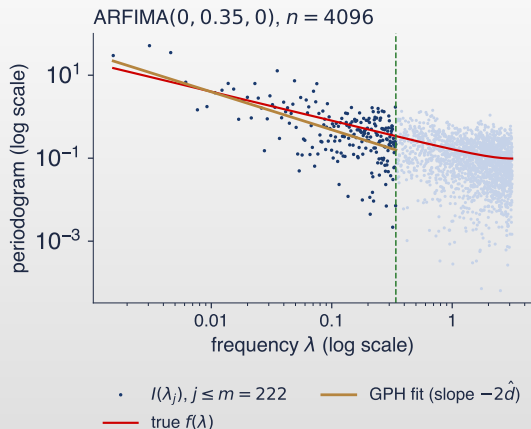
## Prerequisites for Today (8/24): the GPH estimator

- **GPH** Geweke and Porter-Hudak (1983): take logs of  $I(\lambda_j) \approx f(\lambda_j) \xi_j$  near zero and run OLS

$$\log I(\lambda_j) = a + d \left[ -\log \left( 4 \sin^2(\lambda_j/2) \right) \right] + u_j, \quad j = 1, \dots, m$$

- ▶  $\xi_j$ : exponential error with mean 1;  $u_j = \log \xi_j - \mathbb{E} \log \xi_j$ ;  $a$ : intercept
- ▶  $\hat{d}$ : the OLS slope; regressor  $\approx -2 \log \lambda_j$ , so the log-log slope is  $-2d$
- Asymptotic standard error
  - ▶  $\text{Var} u_j = \pi^2/6$  and the regressor has variance  $\approx 4$ :  $\text{SE}(\hat{d}) = \pi/\sqrt{24m}$
  - ▶  $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \pi^2/24)$ ; 95% interval  $\hat{d} \pm 1.96 \text{ SE}$
- Precision grows only with  $\sqrt{m}$ , not with  $\sqrt{n}$ : estimating  $d$  needs long samples

## Prerequisites for Today (9/24): the log-periodogram near zero



- Simulated ARFIMA(0, 0.35, 0),  $n = 4096$
- The  $m = n^{0.65} = 222$  lowest ordinates (dark) lie around a line
  - ▶ GPH:  $\hat{d} = 0.46$ ; local Whittle: 0.44
  - ▶ both within two SEs of  $d = 0.35$
- The scatter around the line is the exponential noise  $\xi_j$ : it does not shrink with  $n$
- Far from zero (pale points) the short-run dynamics bend  $f$ : they must stay out of the regression

## Prerequisites for Today (10/24): the local Whittle estimator

- Local Whittle Robinson (1995b): Gaussian likelihood of  $f(\lambda_j) = G\lambda_j^{-2d}$  on the  $m$  lowest frequencies

$$\hat{d} = \arg \min_d R(d), \quad R(d) = \log \hat{G}(d) - \frac{2d}{m} \sum_{j=1}^m \log \lambda_j, \quad \hat{G}(d) = \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j)$$

- $\hat{G}(d)$ : the level  $G$  that fits best for a given  $d$ ;  $R(d)$ : the profiled objective (a QLIKE-type fit)

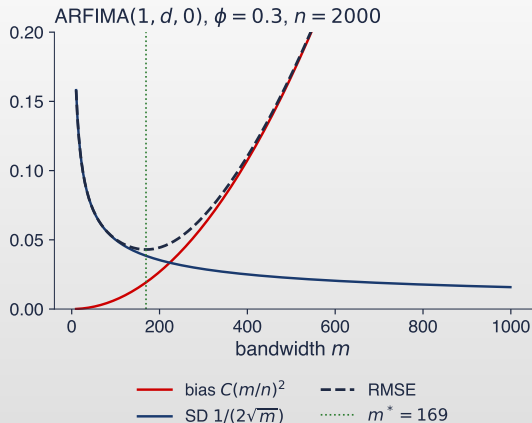
- Inference

- $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$ :  $SE = 1/(2\sqrt{m})$ , smaller than GPH at the same  $m$
  - relative efficiency: the ratio of the two asymptotic variances,  $(\pi^2/24)/(1/4)$
  - consistent for  $-1/2 < d \leq 1$ ; for  $d > 1$  it converges to 1

- Exact local Whittle (ELW) Shimotsu and Phillips (2005), Shimotsu (2010)

- replaces  $\lambda_j^{2d} I(\lambda_j)$  by the periodogram of  $(1 - L)^d(X_t - \hat{\mu})$ ,  $\hat{\mu}$ : estimated mean
  - $N(d, 1/(4m))$  for any  $d$  in the search range, also  $d \geq 1/2$  (nonstationary)

## Prerequisites for Today (11/24): bias and the choice of bandwidth



- Short-run dynamics bias  $\hat{d}$ ; for ARFIMA(1,  $d$ , 0)

$$\mathbb{E}\hat{d} - d \approx C \left( \frac{m}{n} \right)^2, \quad C = \frac{2\pi^2}{9} \cdot \frac{2\phi}{(1-\phi)^2}$$

- $\phi > 0$ : positive bias, growing with  $m$

- MSE = bias<sup>2</sup> +  $\frac{1}{4m}$ ; minimised at

$$m^* = \left( \frac{n^4}{16C^2} \right)^{1/5}$$

- MSE: mean squared error;  $m^*$  grows like  $n^{4/5}$
- chart:  $\phi = 0.3$ ,  $n = 2000$ :  $C = 2.69$ ,  
 $m^* = 169 = n^{0.68}$

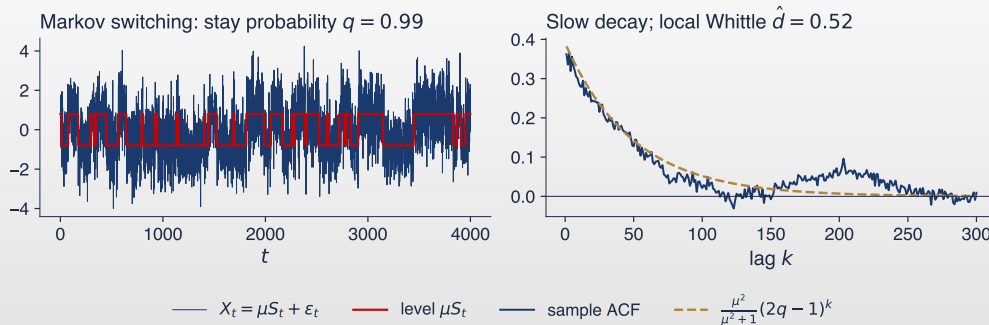
## Prerequisites for Today (12/24): level shifts that mimic long memory

- **Two-state Markov chain**  $S_t \in \{-1, 1\}$ : it stays with probability  $q$ , switches with  $1 - q$

$$X_t = \mu S_t + \varepsilon_t, \quad \mathbb{E}(S_t S_{t+k}) = (2q - 1)^k$$

- ▶  $\mu$ : size of the level;  $\varepsilon_t$ : i.i.d.  $N(0, 1)$ , independent of  $S$ ; expected duration of a regime  $1/(1 - q)$
- ▶ the ACF decays exponentially: short memory in the strict sense
- **Half-life** of a geometric decay  $\rho(k) = \rho(0) r^k$ 
  - ▶  $k_{1/2} = \ln 0.5 / \ln r$ : the lag at which the autocorrelation has halved
- Rare shifts, structural breaks and slowly varying trends put power at the lowest frequencies [Diebold and Inoue \(2001\)](#)
  - ▶  $\hat{d}$  is then large although the process has short memory: **spurious long memory**

## Prerequisites for Today (13/24): a simulated switching series



- $q = 0.99$  (regimes last 100 periods on average),  $\mu = 0.8$ ,  $n = 4000$
- Local Whittle with  $m = n^{0.65}$  gives  $\hat{d} = 0.52$ , a value typical of a long-memory series
- Over a few hundred lags the slow geometric decay cannot be told apart from a hyperbolic one

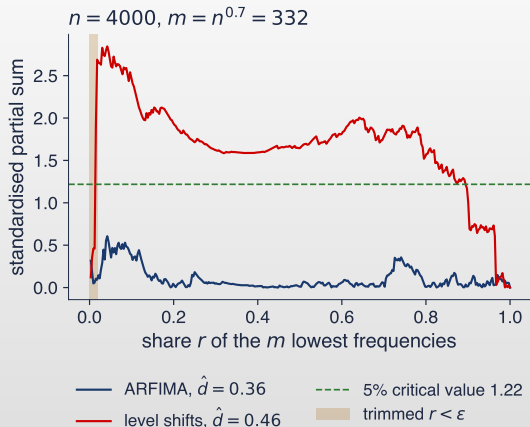
## Prerequisites for Today (14/24): the Qu test

- **Qu test** [Qu \(2011\)](#):  $H_0$  stationary long memory,  $f(\lambda) = G\lambda^{-2d}$  on the  $m$  lowest frequencies

$$W = \sup_{r \in [\varepsilon, 1]} \left( \sum_{j=1}^m \nu_j^2 \right)^{-1/2} \left| \sum_{j \leq mr} \nu_j \left( \frac{\lambda_j^{2\hat{d}} I(\lambda_j)}{\hat{G}} - 1 \right) \right|$$

- ▶  $\hat{d}$ ,  $\hat{G} = \hat{G}(\hat{d})$ : local Whittle estimates;  $\nu_j = \log \lambda_j - \frac{1}{m} \sum_i \log \lambda_i$
  - ▶ the bracket is the score of local Whittle at frequency  $j$ ; its full sum is zero at  $\hat{d}$
  - ▶  $r$ : share of the  $m$  frequencies summed;  $\varepsilon$ : trimming, the first  $\varepsilon m$  partial sums are skipped
- **Reading**
- ▶ under  $H_0$  the partial sums stay small at every  $r$ ; level shifts make the lowest ordinates too large: a big  $W$
  - ▶ reject for  $W$  above the critical value; with  $m = n^{0.7}$ ,  $\varepsilon = 0.02$  (simulated): 1.09 (10%), 1.22 (5%), 1.56 (1%)

## Prerequisites for Today (15/24): the Qu partial sums



- Two simulated series,  $n = 4000$
- ARFIMA(0, 0.35, 0): the curve stays low
  - ▶  $W = 0.61$ : do not reject
- Markov level shifts ( $q = 0.997, \mu = 0.8$ )
  - ▶  $\hat{d}$  looks like long memory
  - ▶ but the curve jumps at small  $r$ :  $W = 2.85$ , reject at 1%
- A rejection says only that stationary long memory does not fit: breaks, shifts or trends are alternatives

## Prerequisites for Today (16/24): fractional cointegration

- **Fractional cointegration:**  $p$  series, each with memory  $d$ , whose combinations  $\beta'X_t$  have memory  $d - b$ ,  $b > 0$
- **FCVAR Johansen and Nielsen (2012):** a VECM written in fractional differences

$$\Delta^d X_t = \alpha \beta' L_b \Delta^{d-b} X_t + \sum_{i=1}^k \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t, \quad \Delta^d = (1 - L)^d, \quad L_b = 1 - \Delta^b$$

- ▶  $\beta$  ( $p \times r$ ): cointegrating vectors;  $\alpha$  ( $p \times r$ ): adjustment speeds;  $r$ : cointegration rank
- ▶  $L_b$ : fractional lag operator (only past values);  $k$ : number of lagged short-run terms,  $\Gamma_i$  their matrices;  $b = d = 1$ : the usual VECM
- Estimation and rank tests
  - ▶ for fixed  $(d, b)$ : Johansen reduced-rank regression; profile likelihood: maximise over a grid of  $(d, b)$
  - ▶ trace test of rank  $r$  against  $p$ :  $LR = 2(\ell_p - \ell_r)$ ,  $\ell$ : maximised log likelihoods;  $\chi^2_{(p-r)^2}$  when  $b < 1/2$

## Prerequisites for Today (17/24): fractional Brownian motion

- **Fractional Brownian motion** (fBm)  $W_t^H$  [Mandelbrot and Van Ness \(1968\)](#): Gaussian,  $W_0^H = 0$ , mean 0, covariance

$$\mathbb{E}(W_t^H W_s^H) = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}), \quad \mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$$

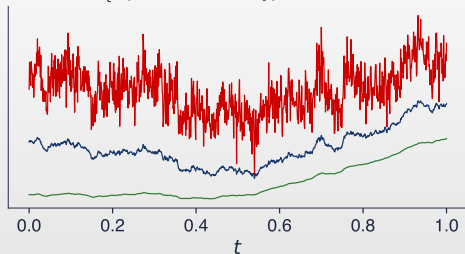
- ▶  $H \in (0, 1)$ : Hurst exponent;  $H = 1/2$ : Brownian motion, independent increments
- ▶  $H < 1/2$ : rough (irregular) paths, negatively correlated increments;  $H > 1/2$ : smooth paths, persistent increments

- **Fractional Gaussian noise** (fGn): the unit increments  $W_{t+1}^H - W_t^H$

$$\gamma_k = \rho(k) = \frac{1}{2} (|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H})$$

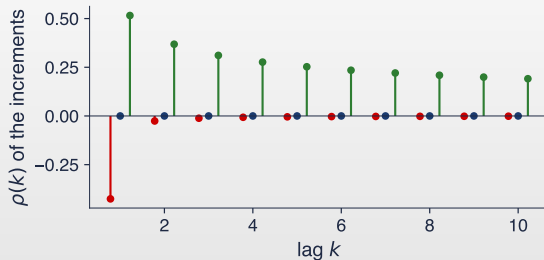
- ▶  $\rho(1) = 2^{2H-1} - 1$ ; for  $H > 1/2$ ,  $\rho(k) \sim H(2H-1)k^{2H-2}$ : long memory with  $d = H - 1/2$

## Prerequisites for Today (18/24): rough and smooth paths

Paths of  $W_t^H$  (shifted vertically)

—  $H = 0.1$     —  $H = 0.5$     —  $H = 0.8$

Fractional Gaussian noise



- Left: fBm paths from the same random numbers for  $H = 0.1, 0.5, 0.8$
- Right:  $H = 0.1$  gives  $\rho(1) \approx -0.43$  and almost nothing beyond: the path reverts at once
- $H = 0.8$ : positive, slowly decaying autocorrelations, the long memory of the previous frames

## Prerequisites for Today (19/24): circulant embedding

- **Goal:** simulate  $n$  points of a stationary Gaussian series with autocovariances  $\gamma_0, \dots, \gamma_{n-1}$  exactly
  - ▶ Cholesky factorisation of the  $n \times n$  covariance costs  $O(n^3)$
- **Davies–Harte** [Davies and Harte \(1987\)](#): embed the covariance in a circulant matrix of size  $M = 2(n - 1)$

$$c = (\gamma_0, \gamma_1, \dots, \gamma_{n-1}, \gamma_{n-2}, \dots, \gamma_1), \quad \lambda_k = \sum_{j=0}^{M-1} c_j e^{-2\pi i j k / M}$$

- ▶ circulant: each row is the previous one shifted by one place;  $c$ : its first row
- ▶  $\lambda_k$ : its eigenvalues, the FFT (fast Fourier transform) of  $c$ , real because  $c$  is symmetric
- ▶ if all  $\lambda_k \geq 0$ :  $Y = \text{FFT}(\sqrt{\lambda/M} Z)$ ,  $Z$  complex Gaussian; the first  $n$  values have the target covariance
- ▶ cost  $O(n \log n)$ ; for fGn all  $\lambda_k \geq 0$  for every  $H$

## Prerequisites for Today (20/24): realised variance and moment scaling

- ▣ **Realised variance** of day  $t$  from  $N_t$  intraday returns  $r_{t,i}$  (e.g. 5-minute)

$$RV_t = \sum_{i=1}^{N_t} r_{t,i}^2, \quad \sigma_t = \sqrt{RV_t}$$

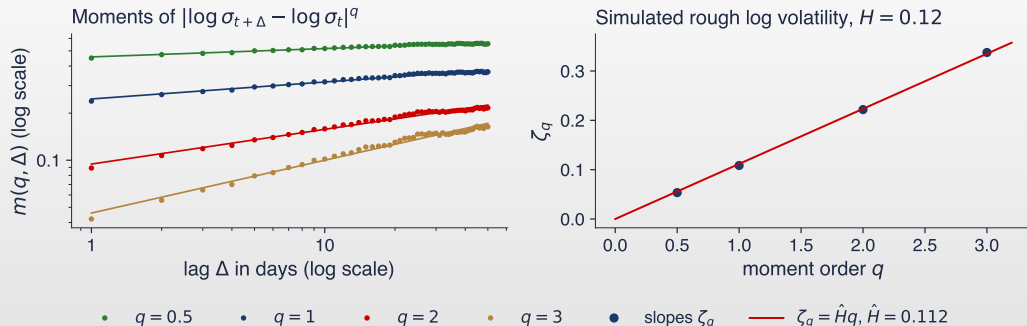
- ▶ a noisy measure of the daily variance; fewer returns per day: a larger measurement error

- ▣ **Moment scaling** Gatheral, Jaisson and Rosenbaum (2018): for  $\log \sigma_t = \nu W_t^H$  the moments of the increments are power laws

$$m(q, \Delta) = \frac{1}{N} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q \propto \Delta^{\zeta_q}, \quad \zeta_q = qH$$

- ▶  $q$ : moment order;  $\Delta$ : lag in days;  $N$ : number of pairs;  $\nu$ : volatility of volatility (vol-of-vol)
- ▶  $\zeta_q$ : slope of  $\log m(q, \Delta)$  on  $\log \Delta$ ; then  $\hat{H} = \sum_q q \zeta_q / \sum_q q^2$  (regression through the origin)
- ▶ i.i.d. measurement error of variance  $s^2$  adds an intercept:  $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$

## Prerequisites for Today (21/24): the scaling regression



- Simulated daily log volatility  $0.3 W_t^H$  with  $H = 0.12$ , 4000 days, no measurement error
- Left: straight lines on log scales, one per  $q$ ; right: the slopes lie on a line through the origin,  $\hat{H} = 0.112$
- Departures from linearity in  $q$  (multifractality, heavy-tailed increments) show up at large  $q$

## Prerequisites for Today (22/24): HAR, FIGARCH and lognormal forecasts

- **HAR Corsi (2009)**: daily, weekly and monthly averages of the past,  $y_t = \log RV_t$

$$y_{t+h} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+h}, \quad \bar{y}_t^{(K)} = \frac{1}{K} \sum_{j=0}^{K-1} y_{t-j}$$

- ▶  $h$ : horizon; OLS; a short-memory AR(22) with restricted weights that mimics long memory

- **FIGARCH Baillie, Bollerslev and Mikkelsen (1996)**: the GARCH variance with a fractional difference of the squared shocks  $e_t^2$

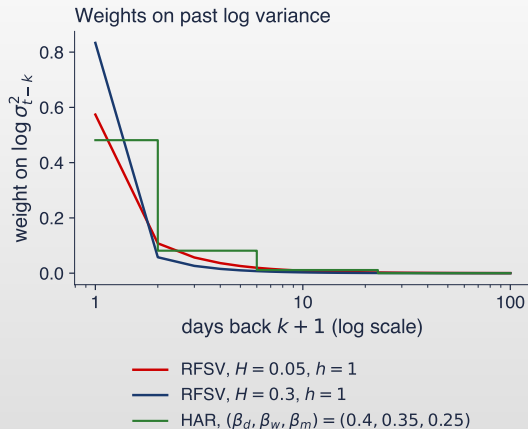
$$\sigma_t^2 = \omega + [1 - (1 - \beta L)^{-1} (1 - \varphi L)(1 - L)^d] e_t^2$$

- ▶  $\omega > 0$ ,  $\beta$ ,  $\varphi$ : GARCH-type coefficients; the bracket is an ARCH( $\infty$ ) filter whose weights sum to one

- **Lognormal correction**: models forecast  $\log RV$ , the target is  $RV$

- ▶ if  $\log RV_{t+h} \sim N(\mu, s^2)$ :  $\mathbb{E}RV_{t+h} = e^{\mu+s^2/2}$ ;  $e^\mu$  alone is the median, too low

## Prerequisites for Today (23/24): the RFSV predictor



- RFSV:  $\log \sigma_t = \nu W_t^H$ ; the forecast is a power-kernel average of the past

$$\widehat{\log \sigma_{t+h}^2} = \sum_k w_k \log \sigma_{t-k}^2$$

- ▶  $w_k \propto \int_k^{k+1} \frac{du}{(u+h)u^{H+1/2}}$ , normalised to sum to 1;  $h$ : horizon

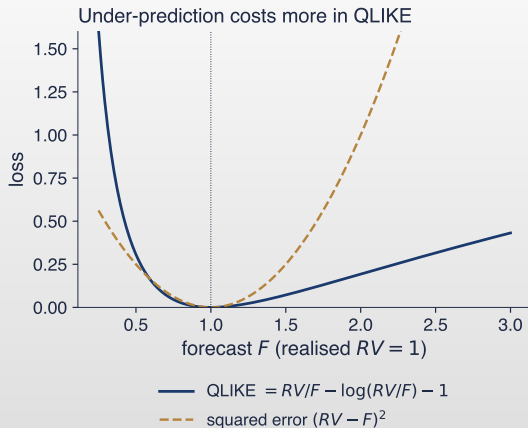
- Variance forecast

$$\exp(\widehat{\log \sigma_{t+h}^2} + 2c\nu^2 h^{2H})$$

- ▶  $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$ ;  $4c\nu^2 h^{2H}$ : the forecast-error variance of  $\log \sigma^2$ , halved as in the lognormal correction

- Chart: the kernel puts most weight on the last day and a long thin tail behind; HAR uses three flat steps

## Prerequisites for Today (24/24): QLIKE and the Diebold–Mariano test



- **QLIKE** [Patton \(2011\)](#) of a variance forecast  $F$

$$\text{QLIKE} = \frac{RV}{F} - \log \frac{RV}{F} - 1 \geq 0$$

- ▶ 0 only if  $F = RV$ ; ranks forecasts correctly although  $RV$  is a noisy proxy
- ▶ penalises under-prediction more (chart)

- **DM test** ([Chapter 1](#)):  $\delta_t$  = loss of model 1 minus loss of model 2

$$DM = \frac{\bar{\delta}}{\sqrt{\hat{\Omega}/T}} \rightarrow N(0, 1)$$

- ▶  $\bar{\delta}$ : mean over  $T$  forecasts;  $\hat{\Omega}$ : Newey–West variance  $\hat{\gamma}_0 + 2 \sum_{k=1}^L (1 - \frac{k}{L+1}) \hat{\gamma}_k$  of  $\delta_t$
- ▶  $DM < -1.96$ : model 1 significantly better at 5%

## A1: ARFIMA autocorrelations and the mean [Solved]

### Task

- ▣ ARFIMA(0, 0.3, 0).
- ▣ 1. Show that  $\rho(1) = d/(1 - d)$  and compute  $\rho(1)$ ,  $\rho(10)$ ,  $\rho(100)$  with the recursion.
- ▣ 2. Compare  $\rho(100)$  with its asymptotic form.
- ▣ 3. For  $n = 1000$ , compute the ratio of the true SD of  $\bar{X}$  to the i.i.d. formula  $\sqrt{\gamma(0)/n}$  (notebook A1).
- ▣ Report: three numbers; two numbers; one number and one sentence.

### Solution

- ▣ 1.  $\rho(1) = \rho(0) \cdot \frac{0+d}{1-d}$ ;  $\rho(1) = 0.429$ ,  $\rho(10) = 0.173$ ,  $\rho(100) = 0.069$
- ▣ 2. Asymptotic formula: 0.069, practically equal already at  $k = 100$
- ▣ 3. Ratio 7.55: the i.i.d. standard error is 7.55 times too small
- ▣ Interpretation: with  $d = 0.3$ , 1000 observations carry the information of far fewer independent ones; HAC with a fixed bandwidth cannot repair this

## A2: fractional differencing weights [Proposed]

Solution discussed in class

### Task

- ▣  $d = 0.4$ . Model: A1.
- ▣ 1. Compute  $\pi_1, \dots, \pi_5$  of  $(1 - L)^{0.4}$ .
- ▣ 2. Compare  $\pi_{100}$  with its asymptotic form.
- ▣ 3. Compute  $\sum_{k=0}^{1000} \pi_k$ ; what is its limit, and what does the gap mean for a truncated filter?
- ▣ Report: five numbers; two numbers; one number and one sentence.

## A3: standard errors of GPH and local Whittle [Solved]

### Task

- ▣  $n = 5000$ ,  $m = \lfloor n^{0.65} \rfloor$ ; a local Whittle estimate  $\hat{d} = 0.45$ .
- ▣ 1. Compute  $m$  and the asymptotic SE of GPH and of local Whittle.
- ▣ 2. Give the 95% interval for  $d$  and the relative efficiency of LW against GPH.
- ▣ 3. Which  $m$  gives a half-width of 0.05?
- ▣ Report: three numbers; an interval and one number; one number.

### Solution

- ▣ 1.  $m = 253$ ; GPH  $\pi/\sqrt{24m} = 0.0403$ ; LW  $1/(2\sqrt{m}) = 0.0314$
- ▣ 2.  $[0.388, 0.512]$ ; efficiency  $(\pi^2/24)/(1/4) = \pi^2/6 = 1.645$
- ▣ 3.  $1.96/(2\sqrt{m}) \leq 0.05 \Rightarrow m \geq 385$
- ▣ Interpretation: the interval includes  $1/2$ , so stationarity is not established; precision is bought with bandwidth, and bandwidth costs bias (A4)

## A4: bias and the optimal bandwidth [Proposed]

Solution discussed in class

### Task

- ▣ ARFIMA(1,  $d$ , 0) with  $\phi = 0.5$ ,  $n = 5000$ . Model: A3.
- ▣ 1. Compute the bias constant  $C$ .
- ▣ 2. For  $m = \lfloor n^{0.8} \rfloor$ , compute the bias and the SD of local Whittle.
- ▣ 3. Compute  $m^*$ , its exponent  $\log m^* / \log n$ , and the bias and SD at  $m^*$ .
- ▣ Report: one number; two numbers; four numbers and one sentence.

## A5: fractional Gaussian noise and scaling [Solved]

### Task

- ▣ fGn with  $H = 0.1, 0.3$  and  $0.7$ .
- ▣ 1. Compute the fGn autocorrelations  $\rho(1)$  and  $\rho(2)$  for each  $H$ .
- ▣ 2. Compute  $m(2, 22)/m(2, 1) = 22^{2H}$ : how much more does log volatility move over a month than over a day?
- ▣ 3. Show that  $\zeta_q = qH$  for fBm.
- ▣ Report: six numbers; three numbers; one derivation.

### Solution

- ▣ 1.  $H = 0.1$ :  $-0.426, -0.026$ ;  $H = 0.3$ :  $-0.242, -0.049$ ;  $H = 0.7$ :  $0.320, 0.189$
- ▣ 2.  $1.86, 6.39, 75.75$ : with  $H = 0.1$  the monthly variance of changes is less than twice the daily one
- ▣ 3.  $W_{t+\Delta}^H - W_t^H \sim N(0, \Delta^{2H})$ :  $\mathbb{E}|\cdot|^q = \Delta^{qH} \mathbb{E}|Z|^q$ ; take logs
- ▣ Interpretation: rough log volatility moves a lot from day to day but drifts little: strong negative correlation of daily changes

## A6: the RFSV kernel and the variance correction [Proposed]

Solution discussed in class

### Task

- ▣  $H = 0.1$ ,  $\nu = 0.3$ , horizon  $h = 1$ ; kernel truncated at 500 days. Model: A5.
- ▣ 1. Compute the share of the weight on the last day, the last 5 and the last 22 days (notebook A6).
- ▣ 2. Compute  $c(H)$  and the variance correction for  $h = 1, 5, 22$ .
- ▣ 3. By how much would ignoring the correction understate the variance forecast at  $h = 22$ ?
- ▣ Report: three shares; four numbers; one number.

## A7: circulant embedding by hand [Solved]

### Task

- ▣ Three points of fGn with  $H = 0.3$ :  $\gamma_0, \gamma_1, \gamma_2$ .
- ▣ 1. Compute  $\gamma_1$  and  $\gamma_2$ .
- ▣ 2. Write the first row of the  $4 \times 4$  circulant and its eigenvalues  $\lambda_k$  ( $M = 4$ ).
- ▣ 3. Can the Davies–Harte method be used?
- ▣ Report: two numbers; a row and four numbers; one sentence.

### Solution

- ▣ 1.  $\gamma_1 = \frac{1}{2}(2^{0.6} - 2) = -0.2421$ ,  
 $\gamma_2 = \frac{1}{2}(3^{0.6} - 2 \cdot 2^{0.6} + 1) = -0.0491$
- ▣ 2.  $c = (\gamma_0, \gamma_1, \gamma_2, \gamma_1)$ ;  $\lambda_0 = \gamma_0 + 2\gamma_1 + \gamma_2 = 0.4666$ ,  
 $\lambda_1 = \lambda_3 = \gamma_0 - \gamma_2 = 1.0491$ ,  $\lambda_2 = \gamma_0 - 2\gamma_1 + \gamma_2 = 1.4352$
- ▣ 3. Yes: all eigenvalues are positive, so the circulant is a valid covariance and its first 3 coordinates have the fGn covariance
- ▣ Interpretation: the embedding doubles the size but turns an  $O(n^3)$  factorisation into two FFTs

## A8: switching that mimics long memory [Proposed]

Solution discussed in class

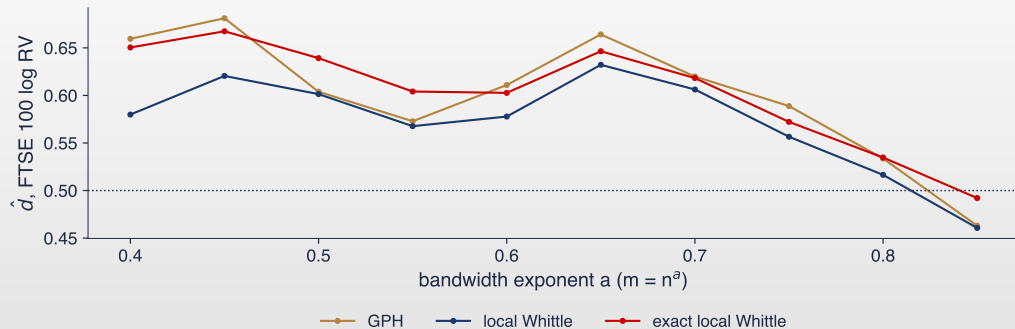
### Task

- $X_t = \mu S_t + \varepsilon_t$ ,  $S_t \in \{-1, 1\}$  a Markov chain that stays with probability  $q = 0.995$ ,  $\mu = 1$ ,  $\varepsilon_t \sim N(0, 1)$ . Model: [A1](#).
- 1. Show that  $\rho(k) = \frac{\mu^2}{\mu^2 + 1} (2q - 1)^k$  and compute  $\rho(1)$ ,  $\rho(100)$ ,  $\rho(500)$ .
- 2. Compute the half-life of the autocorrelation.
- 3. Fit  $\log \rho(k)$  on  $\log k$  for  $k = 5, \dots, 200$  and convert the slope to  $d$ .
- Report: three numbers; one number; one number and one sentence.

## B1: memory of FTSE 100 realised variance [Solved]

- ▣ **Question:** how large is  $d$  for the log realised variance of the FTSE 100, and is it robust to the bandwidth and to level shifts?
- ▣ **Inputs:** Oxford-Man 5-minute RV of the FTSE 100,  $n = 5\,586$  days to 25 February 2022
- ▣ **Tasks**
  1. Estimate  $d$  of log RV by GPH, local Whittle and exact local Whittle for  $m = n^{0.5}, n^{0.6}, n^{0.7}$ , with standard errors.
  2. Plot  $\hat{d}(m)$  for  $m = n^{0.40}, \dots, n^{0.85}$ .
  3. Compute the Qu statistic with  $m = n^{0.7}$  and  $\varepsilon = 0.02$ .
  4. Interpretation: is the memory of FTSE 100 volatility genuine, and is it stationary?
- ▣ **Report:** a table of nine estimates with SEs, one chart, one statistic, one sentence
- ▣ **Notebook:** section [B1](#)

## B1: solution (1/2) [Solved]



## B1: solution (2/2) [Solved]

| $m$ | GPH (SE)      | LW    | ELW   | SE (LW, ELW) |
|-----|---------------|-------|-------|--------------|
| 74  | 0.604 (0.075) | 0.601 | 0.639 | 0.058        |
| 177 | 0.611 (0.048) | 0.578 | 0.603 | 0.038        |
| 419 | 0.620 (0.031) | 0.606 | 0.618 | 0.024        |

- ▣ Qu:  $W = 0.33$  against the 10% value 1.09: no evidence of level shifts
- ▣ Interpretation:  $\hat{d} \approx 0.6$  at every bandwidth with a flat  $\hat{d}(m)$ : genuine long memory, at or just beyond the stationarity border; report ELW

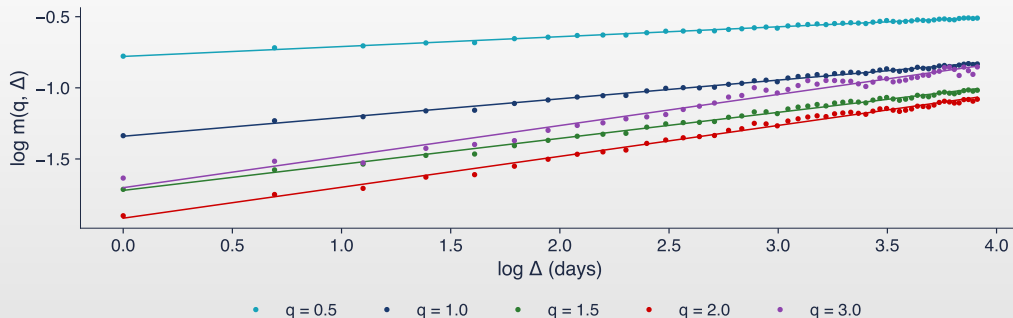
## B2: BET and EUR/RON with the Qu test [Proposed]

- **Question:** is the memory of  $|r_t|$  on the Bucharest market and for the leu genuine, or does it come from regime changes?
- **Inputs:** BET 2000–2026 and from 2010; EUR/RON (BNR) 2005–2026 and from 2014; model: [B1](#)
- **Tasks**
  1. Estimate  $d$  of  $|r_t|$  by local Whittle with  $m = n^{0.65}$  on the full sample and after the cut date.
  2. Compute the Qu statistic ( $m = n^{0.7}$ ,  $\varepsilon = 0.02$ ) on the same samples.
  3. Interpretation: which of the two markets has genuine long memory in volatility?
- **Report:** four estimates with SEs, four statistics, one sentence
- **Notebook:** section [B2](#)

## B3: roughness of the Euro Stoxx 50 [Solved]

- ▣ **Question:** does the Gatheral–Jaisson–Rosenbaum scaling hold for the Euro Stoxx 50, and how rough is its volatility?
- ▣ **Inputs:** Oxford-Man 5-minute RV,  $n = 5\,635$  days;  $\sigma_t = \sqrt{RV_t}$ ;  $q = 0.5, 1, 1.5, 2, 3$ ;  $\Delta = 1, \dots, 50$
- ▣ **Tasks**
  1. Compute  $m(q, \Delta)$  and the slopes  $\zeta_q$ ; estimate  $H$  by regressing  $\zeta_q$  on  $q$  through the origin.
  2. Estimate  $H$  with [the measurement-error intercept](#).
  3. Repeat the first estimate on the two halves of the sample.
  4. Interpretation: is  $\zeta_q$  linear in  $q$ , and what does a departure mean?
- ▣ **Report:** five slopes, three values of  $H$ ,  $\nu$ , one sentence
- ▣ **Notebook:** section [B3](#)

## B3: solution [Solved]



□  $\zeta_q = 0.069, 0.132, 0.183, 0.217, 0.218$ ;  $\hat{H} = 0.093$ ,  $\hat{\nu} = 0.38$ ; with the error intercept 0.109; halves 0.091 and 0.095

□ Interpretation: linear up to  $q = 2$ , flat between  $q = 2$  and 3: large moments are driven by a few extreme days (heavy tails of the increments), so  $H$  from  $q \leq 2$  is the safer reading; roughness is stable over time

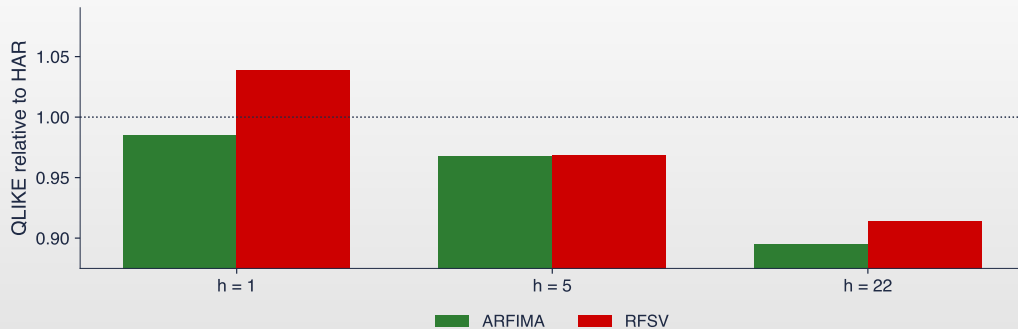
## B4: Bitcoin roughness by sampling interval [Proposed]

- **Question:** does the estimate of  $H$  for Bitcoin depend on the sampling interval of the realised variance?
- **Inputs:** Binance RV of Bitcoin from 1-, 5-, 15-, 30- and 60-minute returns, 2018–2026; model: [B3](#)
- **Tasks**
  1. Estimate  $H$  for each RV by the moment regression and with the error intercept.
  2. Estimate  $d$  of log RV by local Whittle with  $m = n^{0.65}$ .
  3. Interpretation: which direction does a coarser grid push  $H$ , and why?
- **Report:** ten values of  $H$ , five values of  $d$ , one sentence
- **Notebook:** section [B4](#)

## B5: forecasting FTSE 100 realised variance [Solved]

- **Question:** does a rough or a long-memory model forecast FTSE 100 realised variance better than HAR?
- **Inputs:** rolling window of 1000 days, re-estimation every 20 days; out of sample 30 December 2003 – 26 January 2022 ( $T = 4\,565$ ); horizons 1, 5, 22 days
- **Tasks**
  1. Forecast  $RV_{t+h}$  with HAR, ARFIMA(0,  $d$ , 0) and RFSV, with the lognormal corrections.
  2. Compute the average QLIKE of each model and the ratio to HAR.
  3. Compute the DM statistics against HAR.
  4. Interpretation: which model would you use at each horizon?
- **Report:** nine losses, six ratios, six DM statistics, one sentence
- **Notebook:** section [B5](#)

## B5: solution (1/2) [Solved]



## B5: solution (2/2) [Solved]

| $h$ | QLIKE HAR | ARFIMA / HAR | DM    | RFSV / HAR | DM    |
|-----|-----------|--------------|-------|------------|-------|
| 1   | 0.230     | 0.985        | -0.98 | 1.038      | 1.49  |
| 5   | 0.324     | 0.968        | -1.94 | 0.968      | -1.33 |
| 22  | 0.561     | 0.895        | -2.12 | 0.914      | -1.93 |

- Interpretation: HAR at one day (RFSV is worse); at 22 days both ARFIMA ( $p = 0.034$ ) and RFSV ( $p = 0.053$ ) beat HAR by about a tenth; median rolling  $\hat{H} = 0.11$

## B6: forecasting CAC 40 realised variance [Proposed]

- **Question:** does the FTSE 100 ranking carry over to the CAC 40?
- **Inputs:** the same design, out of sample 22 December 2003 – 26 January 2022 ( $T = 4\,631$ ); model: [B5](#)
- **Tasks**
  1. Repeat the three forecasts and the QLIKE ratios.
  2. Compute the DM statistics against HAR.
  3. Interpretation: is there a single best model across the two indices?
- **Report:** six ratios, six DM statistics, one sentence
- **Notebook:** section [B6](#)

## C1: co-memory of DAX and CAC 40 volatility [Proposed]

- **Question:** do the volatilities of two neighbouring markets share one fractional trend?
- **Inputs:** log 5-minute RV of the DAX and the CAC 40 (Oxford-Man),  $n = 5\,600$  common days; FCVAR with  $k = 0$ ; model: [B1](#)
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, model, grid of  $(d, b)$ , tests, what would change your mind.
  2. Estimate  $d$  of each series and of the spread  $\beta' X_t$  by local Whittle, then the FCVAR by profile likelihood with trace tests.
  3. Interpretation: what do the data support, and what would you add to the model?
- **Report:** a five-line plan, five estimates, two tests, a project plan
- **Notebook:** section [C1](#)

## C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise long memory and rough volatility. The answer:
- ▣ (a) An ARFIMA with  $d = 0.3$  has autocorrelations that decay exponentially, only more slowly than an AR(1).
- ▣ (b) The standard error of local Whittle is  $\pi/\sqrt{24m}$ .
- ▣ (c) A larger bandwidth  $m$  is always better, because the variance falls.
- ▣ (d) If the Qu test rejects, the series has structural breaks.
- ▣ (e)  $H = 0.1$  means that volatility has short memory, because  $H < 0.5$  is antipersistence.
- ▣ (f) FIGARCH with  $0 < d < 1$  is covariance stationary with a finite unconditional variance.
- ▣ (g) Exact local Whittle remains consistent for  $d = 1.2$ , whereas local Whittle does not.
- ▣ Tasks
  - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - ▶ 2. Report: a list of seven verdicts with one line of justification each.

## Key takeaways

- ▣ Long memory makes i.i.d. and HAC standard errors far too small
- ▣ Local Whittle is the default; exact local Whittle when  $d$  may exceed  $1/2$ ; always vary  $m$
- ▣ Test against level shifts before claiming long memory
- ▣ Log volatility is rough ( $H \approx 0.1$ ) and persistent; the proxy affects  $H$
- ▣ Forecast gains over HAR are small and horizon-dependent

## After the seminar

- Lecture 10 develops each topic of today: spectral theory, asymptotics of the estimators, the Qu test, FCVAR, FIGARCH, rough volatility and its forecasts
- C1 can grow into a team project: co-memory of Central and Eastern European volatilities
- Reading: [Robinson \(1995b\)](#); [Shimotsu and Phillips \(2005\)](#); [Qu \(2011\)](#); [Gatheral, Jaisson and Rosenbaum \(2018\)](#)
- **The seminar is for practice and is not graded**

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